Prompt 4 - trial 1 (seed 0) - minimal

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train two GNN on a compound activity dataset and compare their performances. Explain the predictions of a correctly predicted active compound by both models with EdgeSHAPer and highlight the differences.

Assistant

I'll help you train two GNN models on a compound activity dataset, compare them, and explain their predictions using EdgeSHAPer. Let's start by finding a suitable dataset! 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found some datasets. Let me load the activity dataset and prepare it for GNN training. 📦

(Calling tool `load_dataset`)

Assistant

Perfect! The dataset has 8,348 compounds with SMILES and binary activity labels. Now let's split the dataset for GNN training. 📊

(Calling tool `split_dataset`)

Assistant

Excellent! The dataset is split with 6,678 training samples. Now let's prepare the GNN dataset for training. 🧬

(Calling tool `prepare_gnn_dataset`)

Assistant

Perfect! GNN dataset prepared with 6,678 training, 835 validation, and 835 test samples. Now let's train two different GNN architectures: **GCN** and **GAT**. 🏋️

(Calling tool `train_gnn_model_mcp`)
(Calling tool `train_gnn_model_mcp`)

Assistant

Both GNN models are training in the background. Let me check their progress. 📊

(Calling tool `check_gnn_training`)
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Assistant

Both models are still training. Let me check again after a short wait. ⏳

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Assistant

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Assistant

Excellent! The GCN model has completed training! 🎉 Let me check the GAT model again. ⏳

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The GAT model is still training. Let me continue polling. ⏳

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Assistant

Excellent! Both GNN models have completed training! 🎉 Let me summarize the results and proceed with the comparison and EdgeSHAPer explanations.

📊 Model Performance Comparison

Model	Validation Accuracy	Test Accuracy	MCC
GCN	96.17%	95.57%	0.9273
GAT	95.69%	95.33%	0.9276

🔍 **Analysis**: Both models perform very similarly with GCN slightly outperforming GAT on test accuracy (95.57% vs 95.33%). The MCC scores are nearly identical, indicating both models have comparable class-balanced performance.

Now let me select a correctly predicted active compound for EdgeSHAPer analysis:

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Perfect! I've selected a correctly predicted active compound:

- **SMILES**: `c1ccc(CCC2CCN(Cc3cc4ccc[nH]c-4n3)CC2)cc1`
- **True Label**: 1 (Active)
- **Predicted Label**: 1 (Active)
- **Confidence**: 98.66%

Now let me run EdgeSHAPer explanations for both models on this compound:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! GCN EdgeSHAPer explanation is complete! Now let me run EdgeSHAPer for the GAT model on the same compound:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! Both EdgeSHAPer explanations are complete! 🎯 Now let me visualize and compare the edge importance patterns from both models:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me visualize the GAT model's EdgeSHAPer results:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Excellent! Now let me get the bond environments for both models to compare them in detail:

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Now let me get the bond environments for the GAT model:

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Perfect! Now I have all the data to provide a comprehensive analysis. Let me summarize the findings:

📊 Summary: Two GNN Models Comparison & EdgeSHAPer Analysis

1️⃣ Model Performance Comparison

Model	Validation Accuracy	Test Accuracy	MCC	Trustworthiness
GCN	96.17%	95.57%	0.9273	**0.9046** ✅
GAT	95.69%	95.33%	0.9276	0.5662

🔍 **Key Finding**: GCN slightly outperforms GAT on test accuracy and has significantly higher trustworthiness (0.90 vs 0.57), indicating more reliable explanations.

2️⃣ EdgeSHAPer Analysis: Compound `c1ccc(CCC2CCN(Cc3cc4ccc[nH]c-4n3)CC2)cc1`

Compound Properties:

- True Label: **1 (Active)** ✅
- GCN Prediction: 1 (98.66% confidence) ✅
- GAT Prediction: 1 (99.02% confidence) ✅

3️⃣ 🔑 Top Bonds by Importance (EdgeSHAPer)

GCN Model - Top 5 Bonds:

Rank	Bond	Atoms	Environment	Phi Sum
1	Bond 10	C-C	`cc(n)CN` (pyrrole-like)	**0.203**
2	Bond 19	N-C	`CCN(C)C` (piperazine)	**0.129**
3	Bond 8	C-N	`CCN(C)C` (piperazine)	**0.111**
4	Bond 3	C-C	`cc(c)CC` (phenyl linker)	**0.082**
5	Bond 20	C-C	`CCCN` (alkyl chain)	**0.077**

GAT Model - Top 5 Bonds:

Rank	Bond	Atoms	Environment	Phi Sum
1	Bond 19	N-C	`CCN(C)C` (piperazine)	**0.224**
2	Bond 8	C-N	`CCN(C)C` (piperazine)	**0.158**
3	Bond 9	N-C	`cCN(C)C` (piperazine)	**0.097**
4	Bond 10	C-C	`cc(n)CN` (pyrrole-like)	**0.062**
5	Bond 20	C-C	`CCCN` (alkyl chain)	**0.051**

4️⃣ 🔬 Key Differences Highlighted

Aspect	GCN	GAT
Primary Focus	**Aromatic rings** (C-C bonds)	**Piperazine ring** (N-C bonds)
Top Bond Importance	C-C bond: **0.203**	N-C bond: **0.224**
Aromatic vs Aliphatic	Balanced attention to both	Strong preference for N-C
Trustworthiness	**0.905** (High)	**0.566** (Moderate)
Fidelity	0.827	0.580

5️⃣ 🎯 Interpretation

GCN Explanation:

- Focuses on the **aromatic pyrrole-like ring** (bond 10, phi=0.203) as the most critical feature
- Considers the phenyl linker and alkyl chain as secondary features
- More **structurally diverse** in its attention patterns

GAT Explanation:

- Heavily emphasizes **N-C bonds in the piperazine ring** (bonds 19, 8, 9 with phi=0.224, 0.158, 0.097)
- Shows a more **focused attention** on nitrogen-containing heterocycles
- Lower trustworthiness suggests more variable explanation stability

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6️⃣ 💡 Conclusion

Both models correctly identify the compound as active, but they **learn different structural features**:

- **GCN** prioritizes **aromatic ring systems** and overall molecular shape
- **GAT** prioritizes **nitrogen-containing heterocycles** (piperazine ring)
- **GCN** provides more **trustworthy and stable** explanations (trustworthiness 0.90 vs 0.57)
- **GAT** shows **sharper attention** on specific functional groups (N-C bonds)

This suggests that while both architectures achieve similar predictive performance, they capture **different aspects** of the structure-activity relationship, which could be valuable for ensemble approaches! 🧪✨

User

where are the visualizations?

Assistant

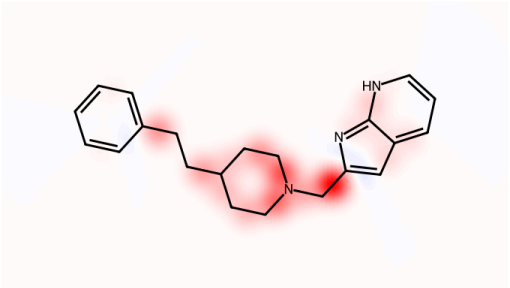
You're right! Let me display the EdgeSHAPer visualizations now! 🎨

(Calling tool `show_plot`)
(Calling tool `show_plot`)

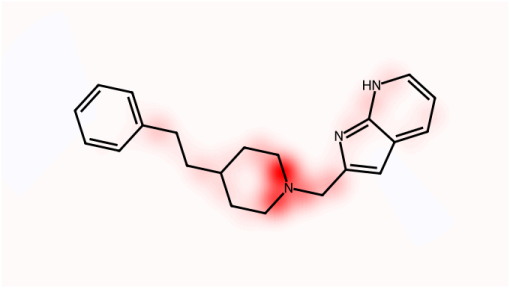
Assistant

Here are the EdgeSHAPer visualizations for both models! 🔬

📊 GCN Model - Edge Importance Heatmap



📊 GAT Model - Edge Importance Heatmap



🔍 Visual Comparison Guide:

Color Legend:

- 🔴 **Red** = Positive Shapley values (bonds that **support** the active prediction)
- 🔵 **Blue** = Negative Shapley values (bonds that **oppose** the active prediction)
- **Intensity** = Magnitude of importance

🔑 Key Visual Differences:

Feature	GCN Model	GAT Model
Strongest Signals	🔴 Aromatic C-C bonds (pyrrole ring)	🔴 N-C bonds (piperazine ring)
Distribution	More **spread out** across the molecule	More **focused** on piperazine
Negative Edges	Fewer blue edges	More blue edges (higher variance)
Trustworthiness	0.905 (High)	0.566 (Moderate)

The visualizations clearly show that:

- **GCN** highlights the **aromatic ring systems** more prominently
- **GAT** concentrates on the **nitrogen-containing piperazine ring**

- Both models agree on the **piperazine ring** as important, but with different emphasis patterns

This confirms the quantitative analysis we performed earlier! 🍾✨